The supplemental materials for paper: A Bio-Inspired Energy Cell Framework for Developing Flexible and Resilient Urban Energy Systems

I. A Comprehensive Comparison Among the EC And Related Concepts.

Table S1. The comparison between the proposed EC framework and exiting architecture.

Architecture	Basic Unit	Covered Sectors	Boundary Characteristic	Operational Mode	Typical Application Scenario
Energy Hub	Energy-coupling devices	Multi-energy system	Single-site, without an explicit inter-unit boundary	Device-level dispatch	Optimal multi-energy dispatch at a single site
Microgrid	Local grid	Mainly power system	Relatively fixed, usually defined by physical connection or islanding capability	Islanded/grid-connected operation	Local energy autonomy and islanded operation
Cellular Grid	Grid cell	Power system	Fixed, partitioned by operational boundaries	Distributed coordination	Voltage and frequency control
Urban Cell	Functionalized urban block	Energy-society system	Fixed, partitioned by urban blocks	Planning-oriented	Urban planning and design
Proposed EC framework	Energy cell	Multi-energy system	Flexible, with dynamic aggregation into EC clusters	Hierarchical coordination across EC-cluster-UES	Economic operation, flexibility utilization, and resilience enhancement

As shown in Table S1, the proposed EC framework differs from several closely related architectures. Compared with the Energy Hub, which mainly focuses on the optimal dispatch of energy-coupling devices at a single site, the EC is not limited to device-level energy conversion, but serves as a distributed multi-energy functional unit embedded in a broader system architecture. Compared with the Microgrid, which mainly emphasizes local autonomy and islanded/grid-connected operation within a relatively fixed boundary, the EC further highlights local multi-energy self-balancing, flexible boundary interaction, and dynamic aggregation with neighboring cells. Compared with the Cellular Grid, which is primarily oriented to distributed coordination in the power sector, the proposed EC extends the cell-based idea to a more general urban multi-energy system context. Compared with the Urban Cell, which is mainly planning- and design-oriented at the spatial scale, the EC framework places stronger emphasis on operational coordination, flexibility utilization, and resilience enhancement.

II. Biological Term Explanations

A. A detailed explanation of Fig.1

In Fig.1, at the component level, representative biological organelles are mapped to key energy-system components: the nucleus corresponds to the energy management center, the vacuole to energy storage, the chloroplast to energy generation, and the mitochondrion to energy utilization. This level emphasizes the functional correspondence between the internal constitution of a biological cell and that of an EC. Specifically,

- Cell chloroplast and EC generation-type units: The compatibility between cell chloroplast and generation-type units in EC is not merely that both are related to energy production, but that both serve as the interface through which external energy is captured and converted into a usable internal energy form. In biological cells, the chloroplast transforms environmental energy into usable biochemical energy; similarly, generation-type units in the EC, especially renewable generation units, convert external primary energy into usable energy.

- Cell vacuole and EC storage-type units: Both cell vacuole and EC storage-type unit have role of buffering, temporary storage, and internal regulation. A vacuole is an internal regulator that helps maintain the stability of the cell. Likewise, storage-type units in the EC are buffering components that shift energy across time, absorb fluctuations, and support local self-balancing.
- Cell mitochondrion and EC consumption-type units: This analogy can be understood from the perspective of functional endpoint of energy utilization. The mitochondrion is where energy substrates (such as the glucose) are processed into ATPs to sustain the activity of the cell; similarly, consumption-type units and local loads in the EC represent the part of the system where energy is ultimately utilized to maintain local services and functions.

At the functional level, the analogy is further extended from component correspondence to system-operating principles.

- The hierarchical organization from cell to organ/tissue to organism-level system is used to motivate the EC-cluster-UES layered structure, highlighting modular aggregation from local units to the whole system.
- The signal interaction among biological cells, such as bioelectrical and synaptic communication, is used to inspire inter-EC information exchange, which may be realized through discrete pulse-like communication rather than continuous global broadcasting.
- The biological regulation and immune-response mechanisms activated under virus invasion or abnormal cells, involving hormones, T-cells, B-cells, are used to analogize the cooperative suppression of uncertainty and disturbances in the EC framework, where neighboring ECs can provide coordinated support through storage and other flexible resources.
- The adaptive metabolic switching of cells under low-oxygen conditions, where ATP production shifts from the mitochondrion-dominated aerobic pathway to glycolysis, is used to analogize adaptive energy substitution in ECs, such as the transition from photovoltaic generation to energy-storage discharge under low solar-radiation conditions.

B. Quick reference index for biological terms & engineering mechanisms.

Biological Item	Biological Meaning	Engineering Mechanism	Functional Similarity
Organismic Cell	The basic functional unit of a living organism, capable of maintaining essential functions through internal regulation	Energy Cell, as the fundamental structural and functional unit of UES, integrating distributed generation, loads, storage, and flexible resources	Both are basic functional units that sustain local operation while contributing to the whole system
Cell Division	The formation of multiple cells from a stem cell	Partitioning the UES into multiple initial ECs based on multi-energy balance and resource allocation	Both create manageable local units from a larger system to support organized operation
Cell Differentiation	Cells specialize into different types according to their roles and characteristics	Functional specialization of ECs according to physical roles, resource endowment, and energy characteristics	Both enable heterogeneous units to perform designated functions within a coordinated system
Cell Nucleus	The control center that regulates cellular behavior and decision-making	Control layer of EC for local energy management and decision-making based on sensed states and interaction signals	Both serve as the decision core of the local unit
Cell Cytoplasm	The internal medium containing functional components that support cellular activity	Physical layer of EC, including energy devices, storage, and aggregated energy facilities	Both contain the physical components that realize the unit's actual functions
Cell Membrane	The interface that separates the cell from its environment and regulates exchanges	Interface layer of EC, supporting standardized information exchange and boundary interaction with neighboring ECs or upper-level operators	Both define boundaries and regulate interaction with the outside environment

Cell-Tissue-System	Multi-level biological organization from local units to larger coordinated structures	EC-cluster-UES hierarchy from local ECs to EC clusters and the whole urban energy system	Both represent hierarchical organization from basic units to system-level coordination
Hormone Regulation	Biological regulation through signaling substances that guide responses across scales	Hierarchical coordination signals used to guide EC scheduling and cross-layer adjustment	Both provide regulation to guide the function adjustment
Reference Hormones	Guidance signals under normal conditions for coordinated regulation	Day-ahead or medium-term reference signals sent from upper levels to guide EC scheduling plans	Both guide local behavior toward a broader coordinated objective under normal conditions
Emergency Hormones	Urgent biological signals released when abnormal conditions are detected	Emergency requests or imbalance signals sent from ECs to neighbors or clusters when local balancing is insufficient	Both trigger additional support only when local regulation is no longer enough
Cell chloroplast	Organelles responsible for energy production	Generation-type flexible resources within EC	Both are associated with energy production capability
Cell Vacuole	Organelles used for storage and regulation of materials	Storage-type flexible resources within EC	Both provide storage and buffering capability for local regulation
Cell Mitochondrion	Organelles responsible for energy consumption and metabolic conversion	Consumption-type flexible demand within EC	Both are associated with energy use and conversion for sustaining local function
Stress Reactivity	Short-term ability to respond and recover from sudden disruptions	Short-term flexibility response capability of resources/ECs/clusters	Both describe fast response to immediate disturbances
Defensive Capability	Mid-term ability to resist risks and take preventive measures	Mid-term capability of ECs/clusters to resist foreseeable operational risks	Both reflect protective capacity against anticipated threats
Immunization	Long-term adaptability and resistance to repeated or evolving threats	Long-term sustainability and adaptability of flexibility planning against future operational changes	Both emphasize enduring protection and adaptation over long horizons
Viral Invasion	External attack that disturbs normal cellular function	Extreme disturbances such as natural disasters	Both represent external disturbance threatening normal system function
Autophagy	Controlled self-reduction of internal activity to maintain survival under stress	Proactive shedding of non-critical loads	Both sacrifice part of local function to preserve the survival of the larger unit
Apoptosis	Programmed removal of damaged cells to protect the organism	Proactive disconnection of an EC from neighboring ECs to isolate fault propagation	Both isolate damaged parts to prevent wider system harm
Paracrine Signaling	Local signaling among nearby cells to coordinate neighboring responses	Parallel coordinated signaling among neighboring EC clusters during post-disaster recovery	Both enable local cooperative recovery through nearby interaction
Synaptic Pulses	Discrete signals transmitted only when bioelectrical thresholds are reached	Event-triggered, encoded discrete communication among ECs under limited bandwidth conditions	Both reduce communication burden by transmitting only significant state changes

III. Modeling Details

A. Guidance on general modeling template of EC-clusters-USE hierarchy

In this hierarchy, the UES layer is responsible for the overall system objective, such as minimizing the total operating cost of the urban energy system while preserving coordinated performance across regions. The EC-cluster layer serves as an intermediate coordination level, where multiple neighboring ECs are grouped to achieve broader regional energy complementation and to coordinate inter-EC exchanges. The EC layer is the local decision-making level, where each EC schedules its internal multi-energy resources, including local generation, storage, conversion, and controllable demand, to satisfy local energy balance as much as possible. Accordingly, a minimal coordination template can be written as follows.

- At the UES layer, the system operator determines the system-level coordination target:

$$\begin{aligned}&\min_{y_c,z}\sum_{c\in\mathcal{C}}F_c\left(y_c\right)+G\left(z\right)\\&\text{s.t.}\begin{cases}z\in\mathcal{Z}\\y_c\in\mathcal{Y}_c\end{cases}\end{aligned}\tag{1}$$

where, y_c denotes the decision variables associated with cluster c and z is the global decision variables, $F_c\left(\cdot\right)$ represents the cluster-level operating metrics (such as the cluster operation cost), and $G\left(\cdot\right)$ captures the system-level coordination benefits or penalty (such as the energy accessibility or wind curtailment cost), $\mathcal{Z}$ and $\mathcal{Y}_c$ are the corresponding feasible region of global and cluster variables.

- At the EC-cluster layer, each cluster coordinates the operation of its member ECs:

$$\begin{aligned}&\min_{r_i,e_c}\sum_{i\in\mathcal{E}_c}f_i\left(r_i\right)+h_c\left(e_c\right)\\&\text{s.t.}\begin{cases}B_c\left(r_i,e_c\right)=y_c\\D_c(x_i,e_c)=r_i\\x_i\in\mathcal{X}_i\end{cases}\end{aligned}\tag{2}$$

where x_i denotes the local dispatch decision of i-th EC, e_c is the cluster coordinate variables, r_i is the reference signal or coordination instruction for each EC, $f_i\left(\cdot\right)$ is the local EC operation targets and $h_c\left(\cdot\right)$ is the coordination discipline. $B_c\left(\cdot\right)$ couples the EC decisions and coordinated decisions to the cluster-level decision y_c , and $D_c\left(\cdot\right)$ decompose the coordinated decisions into lower EC-level reference signal, $\mathcal{X}_i$ is the feasible operating region of i-th EC.

- At the EC layer, each EC performs local multi-energy scheduling:

$$\begin{aligned}&\min_{x_i}f_i\left(x_i;r_i\right)\\&\text{s.t. }x_i\in\mathcal{X}_i=\left\{x_i\left|g_i(x_i)=0,q_i(x_i)\le 0\right.\right\}\end{aligned}\tag{3}$$

where feasible operating region $\mathcal{X}_i$ is consisted of local equality constraints $g_i\left(\cdot\right)$ and inequality constraints $q_i\left(\cdot\right)$. In the normal-operation multi-energy dispatch setting considered here, these local constraints include the energy-balance equations of different energy carriers, device operating limits, conversion relationships, and local network constraints.

In this way, the proposed hierarchy can be interpreted as a coordination structure in which the UES layer determines the global target, the cluster layer coordinates regional interactions, and the EC layer ensures implementable local dispatch.

B. Detailed decomposition approach and solution framework

To illustrate the decomposition process more clearly, we take the EC-cluster layer as a representative example. At this layer, the coordination problem (2) previous aforementioned can be reformulated into a compact form as follows:

$$\left.\begin{aligned}&\min_{r_i,e_c}\sum_{i\in\mathcal{E}_c}f_i\left(r_i\right)+h_c\left(e_c\right)\\&\text{s.t.}\begin{cases}B_c\left(r_i,e_c\right)=y_c\\D_c(x_i,e_c)=r_i\\x_i\in\mathcal{X}_i\end{cases}\end{aligned}\right\}\xrightarrow[\text{Compact Form}]{\text{Linear Assumption}}\left\{\begin{aligned}&\min_{X_i,\hat{X}_i}\sum_{i\in\mathcal{E}_c}f_i\left(\boldsymbol{X}_i\right)\\&\text{s.t.}\begin{cases}\mathbb{A}_i\boldsymbol{X}_i-\mathbb{B}_i\hat{\boldsymbol{X}}_i=0\\\forall i\in\mathcal{E}\end{cases}\end{aligned}\right.\tag{4}$$

where $\boldsymbol{X}_i$ is the local EC related variables, and $\hat{\boldsymbol{X}}_i$ is the auxiliary variable introduced to represent the coordinated information exchanged within the cluster and the derived variables from the constraints, $\mathbb{A}$ and $\mathbb{B}$ are the related coefficient matrices. Under this reformulation, the original cluster-level coordination problem can be decomposed into multiple local EC subproblems together with a cluster-level auxiliary-variable update. This reformulation is always feasible because the inter-layer consistency constraints $B_c\left(r_i,e_c\right)=y_c$ and $D_c(x_i,e_c)=r_i$ ensure the reference signal

for each EC can be meet by local EC and this reference signal can meet the upper layer target. And the objective of the cluster layer is naturally consisted of local EC objectives, therefore the decomposition is logically.

Specifically, the decomposed subproblems for each EC can be distributed solved through an iterative procedure, which consists of four main steps (take k-th iteration as an example):

- local decision-variable update, where each EC independently solves its own subproblem for X_i

$$\boldsymbol{X}_i^{k+1} = \arg\min_{x_i^k} f_i(\boldsymbol{X}_i) + \frac{\tau_i^k(\boldsymbol{X}_i)}{2}\left\| \boldsymbol{X}_i^k - \hat{\boldsymbol{X}}_i^k + \frac{\boldsymbol{\lambda}_i^k}{\tau_i^k(\boldsymbol{X}_i)} \right\|_2^2 \tag{5}$$

- auxiliary-variable update, where the cluster-level shared variable $\hat{X}_i$ is updated to enforce coordination consistency

$$\hat{\boldsymbol{X}}_i^{k+1} = \arg\min_{\hat{x}^k} \frac{\tau_i^k(\boldsymbol{X}_i)}{2} \sum_{i\in\mathcal{E}_c} \left\| \boldsymbol{X}_i^{k+1} - \hat{\boldsymbol{X}}_i^k + \frac{\boldsymbol{\lambda}_i^k}{\tau_i^k(\boldsymbol{X}_i)} \right\|_2^2 \tag{6}$$

- dual-variable update, where the Lagrange multiplier λ_i associated with the coupling constraint is updated

$$\boldsymbol{\lambda}_i^{k+1} = \boldsymbol{\lambda}_i^k + \tau_i^k(\boldsymbol{X}_i)\left(\boldsymbol{X}_i^{k+1} - \hat{\boldsymbol{X}}_i^{k+1}\right) \tag{7}$$

- adaptive penalty update, where the penalty parameter τ_i is adaptively adjusted to improve convergence.

$$\tau_i^{k+1}(\boldsymbol{X}_i^{k+1}) = g\left(\tau_i^k(\boldsymbol{X}_i)\right) \tag{8}$$

Repeating these steps, the local EC decisions, exchanged coordination variables, and multiplier information can iteratively update until the primal and dual residuals satisfy the stopping criterion.

B. Detailed Models in the case studies.

The compact form:

$$\begin{aligned} &\min_{\boldsymbol{x},\boldsymbol{y}} \sum_{t=1}^{T} \boldsymbol{c}^\top \boldsymbol{x} + \lambda \boldsymbol{y} \\ &\text{s.t.} \mathcal{A}_t \boldsymbol{x} = \mathcal{B}_t, \forall t \in T \\ &\quad \mathbf{A_t}\boldsymbol{x} \le \mathbf{b_t}, \forall t \in T \\ &\quad \boldsymbol{x} \in \mathcal{X} \end{aligned} \tag{9}$$

The objective function:

$$\sum_{t=1}^{T} c_B^\top P_t^B + c_*^\top P_t^* + \lambda^\top P_t^{\mathrm{EC}} \tag{10}$$

where the cost coefficient vectors c_B, c_* correspond to the energy purchase prices and operational costs, λ is the penalty coefficient. The energy purchases P_t^B including power, gas, heat, and hydrogen purchases. The scheduled output vector P_t^* collectively represent the decision variables of all energy devices, and inter-ECs power exchange vector P_t^{EC}.

The energy balance constraints:

$$P_t^B + \sum_{i=1}^{N_{\mathrm{CHP}}} P_{i,t}^{\mathrm{CHP}} + \sum_{i=1}^{N_{\mathrm{WG}}} P_{i,t}^{f} + \sum_{i=1}^{N_{\mathrm{BE}}} P_{i,t}^{\mathrm{BE,D}} + \sum_{i=1}^{N_{\mathrm{FC}}} P_{i,t}^{\mathrm{FC}} = \sum_{i=1}^{N_D} P_{i,t}^{D} + \sum_{i=1}^{N_{\mathrm{EB}}} P_{i,t}^{\mathrm{EB}} + \sum_{i=1}^{N_{\mathrm{BE}}} P_{i,t}^{\mathrm{BE,C}} + \sum_{i=1}^{N_{\mathrm{EL}}} P_{i,t}^{\mathrm{EL}} \tag{11a}$$

$$\sum_{i=1}^{N_{\mathrm{CHP}}} Q_{i,t}^{\mathrm{CHP}} = \sum_{i=1}^{N_D} Q_{i,t}^{D} \tag{11b}$$

$$\sum_{i=1}^{N_{\mathrm{CHP}}} G_{i,t}^{\mathrm{CHP}} + G_t^D = G_t^{\mathrm{buy}} \tag{11c}$$

$$\sum_{i=1}^{N_{\mathrm{CHP}}} H_{i,t}^{\mathrm{CHP}} + \sum_{i=1}^{N_{\mathrm{EB}}} H_{i,t}^{\mathrm{EB}} + \sum_{i=1}^{N_{\mathrm{FC}}} H_{i,t}^{\mathrm{FC}} = H_t^D \tag{11d}$$

$$M_t^{\text{buy}} + \sum_{i=1}^{N_{\text{EL}}} M_{i,t}^{\text{EL}} + \sum_{i=1}^{N_{\text{HE}}} M_{i,t}^{\text{HE,D}} = \sum_{i=1}^{N_{\text{FC}}} M_{i,t}^{\text{FC}} + \sum_{i=1}^{N_{\text{HE}}} M_{i,t}^{\text{HE,C}} \tag{11f}$$

where Eq. (11a) ensures the active power balance, while Eq. (11b)- Eq. (11e) enforce the balance constraints for reactive power, gas, heat, and hydrogen, respectively. P_t^B the purchased power, $P_{i,t}^{\text{CHP}}$ the power output of CHP, $P_{i,t}^f$ the power output of wind generation, $P_{i,t}^{\text{BE,D}}$ the discharging power of energy storage, $P_{i,t}^{\text{FC}}$ the power output of fuel cell (FC), $P_{i,t}^D$ the power demand, $P_{i,t}^{\text{EB}}$ the power consumption of electric boiler (EB), $P_{i,t}^{\text{BE,C}}$ the charging power of energy storage, $P_{i,t}^{\text{EL}}$ the power consumption of electrolyzer (EL). $Q_{i,t}^{\text{CHP}}$ the reactive power output of CHP, $Q_{i,t}^D$ the reactive power demand, $G_{i,t}^{\text{CHP}}$ the gas consumption of CHP, G_t^D the gas demand, G_t^{buy} the purchased gas. $H_{i,t}^{\text{CHP}}$ the heat output of CHP, $H_{i,t}^{\text{EB}}$ the heat output of EB, $H_{i,t}^{\text{FC}}$ the heat output of FC, M_t^{buy} the purchased hydrogen, $M_{i,t}^{\text{EL}}$ the hydrogen output of EL, $M_{i,t}^{\text{HE,D}}$ the discharging flow of hydrogen storage, $M_{i,t}^{\text{FC}}$ the hydrogen consumption of FC, $M_{i,t}^{\text{HE,C}}$ the charging flow of hydrogen storage.

Device constraints of CHP:

$$P_t^{\text{CHP}} = \left(\eta_{\text{CHP}}^{\text{P}} \rho_{\text{G2P}}\right) G_t^{\text{CHP}} \tag{12a}$$

$$H_t^{\text{CHP}} = \left(\eta_{\text{CHP}}^{\text{H}} \rho_{\text{G2P}}\right) G_t^{\text{CHP}} \tag{12b}$$

$$\underline{P}_{\text{CHP}} \le P_t^{\text{CHP}} \le \overline{P}_{\text{CHP}} \tag{12c}$$

$$\underline{Q}_{\text{CHP}} \le Q_t^{\text{CHP}} \le \overline{Q}_{\text{CHP}} \tag{12d}$$

$$-\underline{\Delta P}_{\text{CHP}} \le P_t^{\text{CHP}} - P_{t-1}^{\text{CHP}} \le \overline{\Delta P}_{\text{CHP}} \tag{12e}$$

where Eqs. (12a) and (12b) characterize the conversion of natural gas into electricity and heat based on efficiency parameters η and unit conversion factors ρ. Eq. (12c) imposes ramping constraints to limit abrupt changes in CHP output. Eqs. (12d) and (12e) enforce the operational bounds on active and reactive power outputs, while Eq. (12e) defines the maximum allowable reserve capacities that the CHP unit can offer.

Other devices have the same modeling paradigm, and therefore is omitted here for concise.

Device constraints of BESS:

$$E_t^{\text{BE}} = E_{t-1}^{\text{BE}} + \eta_{\text{BE}}^{\text{C}} P_t^{\text{BE,C}} - P_t^{\text{BE,D}} / \eta_{\text{BE}}^{\text{D}} \tag{13a}$$

$$\underline{E}_{\text{BE}} \le E_t^{\text{BE}} \le \overline{E}_{\text{BE}} \tag{13b}$$

$$\underline{P}_{\text{BE,C}} \le P_t^{\text{BE,C}} \le \overline{P}_{\text{BE,C}} \tag{13c}$$

$$\underline{P}_{\text{BE,D}} \le P_t^{\text{BE,D}} \le \overline{P}_{\text{BE,D}} \tag{13d}$$

Eq. (13a) update the energy level of the BESS. Eqs. (13b) and (13c) impose constraints on the maximum and minimum charging and discharging powers. (No additional constraints are required to prevent simultaneous charging and discharging of BESS, since the objective function in (10) is monotonically increasing with respect to related variables).

Power flow constraints:

$$\mathbf{C} \cdot \mathbf{P}_t^F = \left[\boldsymbol{P}_t^B, \boldsymbol{P}_t^{\text{inj}}\right] \tag{14a}$$

$$\mathbf{C} \cdot \boldsymbol{Q}_t^F = \left[0, \boldsymbol{Q}_t^{\text{inj}}\right] \tag{14b}$$

$$\boldsymbol{P}_t^{\text{inj}} = \mathbf{C}_{\text{CHP}} \boldsymbol{P}_t^{\text{CHP}} + \mathbf{C}_{\text{W}} \widetilde{\boldsymbol{P}}_t^{\text{W}} + \mathbf{C}_{\text{FC}} \boldsymbol{P}_t^{\text{FC}} + \mathbf{C}_{\text{BE}} \boldsymbol{P}_t^{\text{BE,D}} - \mathbf{C}_{\text{D}} \boldsymbol{P}_t^{\text{D}} - \mathbf{C}_{\text{EB}} \boldsymbol{P}_t^{\text{EB}} - \mathbf{C}_{\text{BE}} \boldsymbol{P}_t^{\text{BE,C}} - \mathbf{C}_{\text{EL}} \boldsymbol{P}_t^{\text{EL}} \tag{14c}$$

$$\boldsymbol{Q}_t^{\text{inj}} = \mathbf{C}_{\text{CHP}} \boldsymbol{Q}_t^{\text{CHP}} - \mathbf{C}_{\text{D}} \boldsymbol{Q}_t^{\text{D}} \tag{14d}$$

$$\widehat{\mathbf{C}}^{\top} \boldsymbol{v}_t + v_0 \mathbf{C}_0 = 2\left(\mathbf{R}\boldsymbol{P}_t^{\text{inj}} + \mathbf{X}\boldsymbol{Q}_t^{\text{inj}}\right) \tag{14e}$$

$$\underline{\boldsymbol{v}} \le \boldsymbol{v} \le \overline{\boldsymbol{v}} \tag{14f}$$

Eqs. (14a) and (14b) establish the relationships between branch-level active and reactive power flows and bus-level power injections, which are defined in (14c) and (14d) via the device-to-bus mapping matrix $\mathbf{C}_*$. Eq. (14e) connects the injected

powers to bus voltages by relaxing the phase angle constraints. The squared voltage magnitudes are further bounded within specified limits, as expressed in (14f).

C. Settings for case studies.

The detailed model parameters can be found in the attached Excel file.

In economic dispatch under normal operation, the system diagrams for S1-S3 are illustrated as Fig. S1-S3:

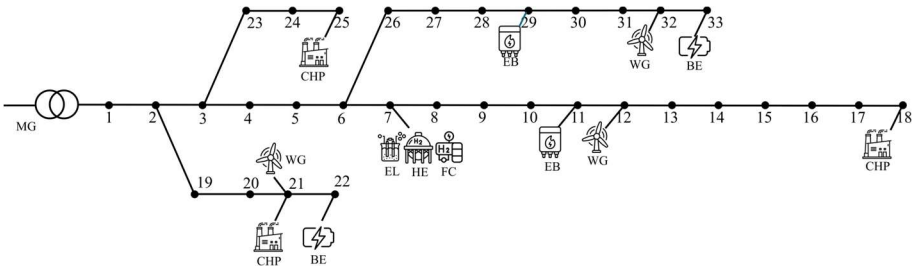

Fig S1. The system diagrams for S1

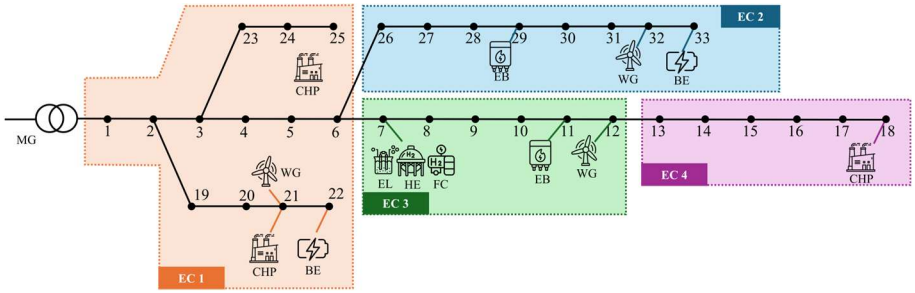

Fig S2. The system diagrams for S2

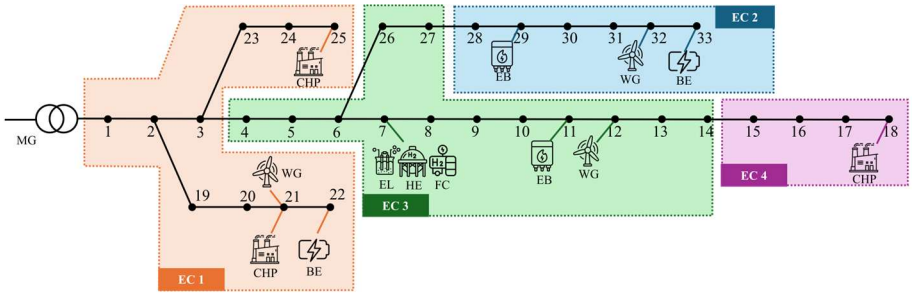

Fig S3. The system diagrams for S3

In resilience enhancement under extreme conditions, the system diagrams without and with flexible boundaries adjustment are illustrated as Fig. S4-S5:

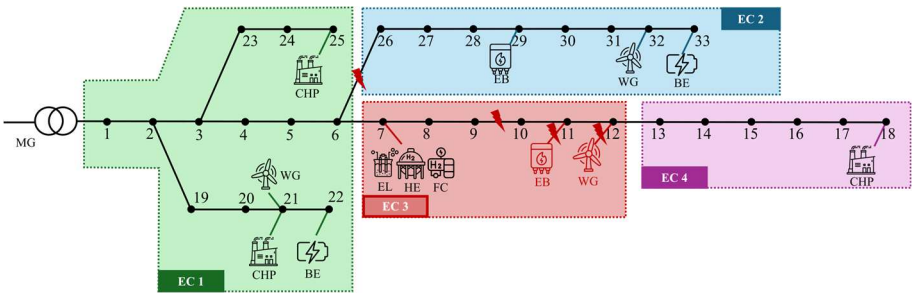

Fig S4. The system diagrams for ECs without boundaries adjustment and clusters

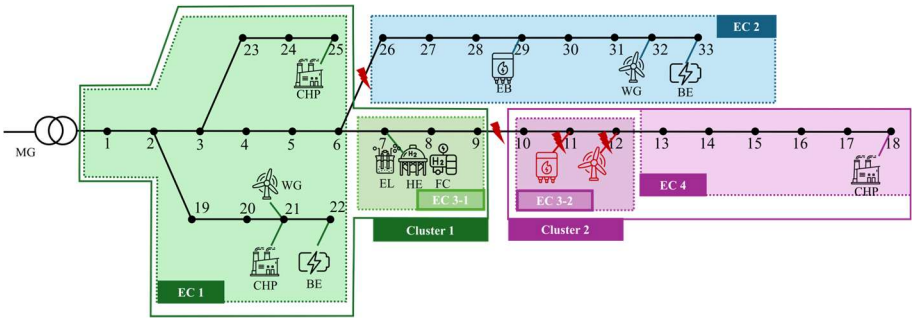

Fig S5. The system diagrams for ECs with boundaries adjustment and clusters

IV. Additional Figures

A. The energy flow diagram in EC

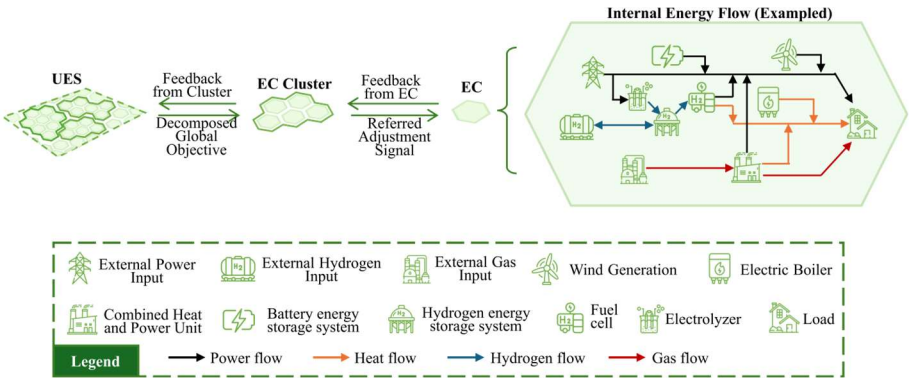

Fig. S6 Energy-flow diagram in an exampled EC and the hierarchy of whole EC framework.